Chapter 10 — Use cases in autonomy and security

Healthcare and finance are not the only high-stakes domains

Learning objectives

- By the end of this part

Autonomous systems data demands

- Self-driving stacks require vast labeled coverage across weather, lighting, traffic
- Logging every dangerous or rare combination on public roads is impractical and costly
- Synthetic environments generate diverse driving clips for navigation, decision-making

Example 10.15 — Pedestrian-in-Rain Edge Case

- Example 10.15 — hands-on module
- A combination rarely present in fleet logs
- Edge-case synthesis improves robustness where real collection is sparse or unsafe
- Explore the chapter example module
- View files: `modules/chapter10/example15/`

Edge cases and safety

- Rare events, jaywalking, sudden obstacles, or unusual road geometry, may never appear
- Synthetic simulation supplies critical scenarios at scale
- Over-reliance remains a risk if generators omit plausible human actions

Augmenting vision and other modalities

- Synthetic data augments sparse or imbalanced real sets across computer vision
- Rendered product photos under varied lighting and clutter diversify backgrounds beyond
- Modified or generated text expands corpora for sentiment and translation tasks

Example 10.16 — Computer-Vision Augmentation

- Example 10.16 — hands-on module
- Example 10.16 shows a vision team rendering synthetic product photos with varied lighting
- Explore the chapter example module
- View files: `modules/chapter10/example16/`

Cybersecurity and evolving threats

- Cyber threats evolve quickly; real attack logs are incomplete and sensitive
- Insider-threat models without waiting for live incidents

Example 10.17 — Ransomware Attack Simulation

- Example 10.17 — hands-on module
- Example 10.17 describes a security operations center replaying synthetic ransomware
- Explore the chapter example module
- View files: `modules/chapter10/example17/`

Takeaways

- Autonomy benefits from synthetic rare-scenario coverage
- All three require validation that synthetic behavior matches plausible real threats and

Next

- Complete the quiz for this part
- Regulatory considerations